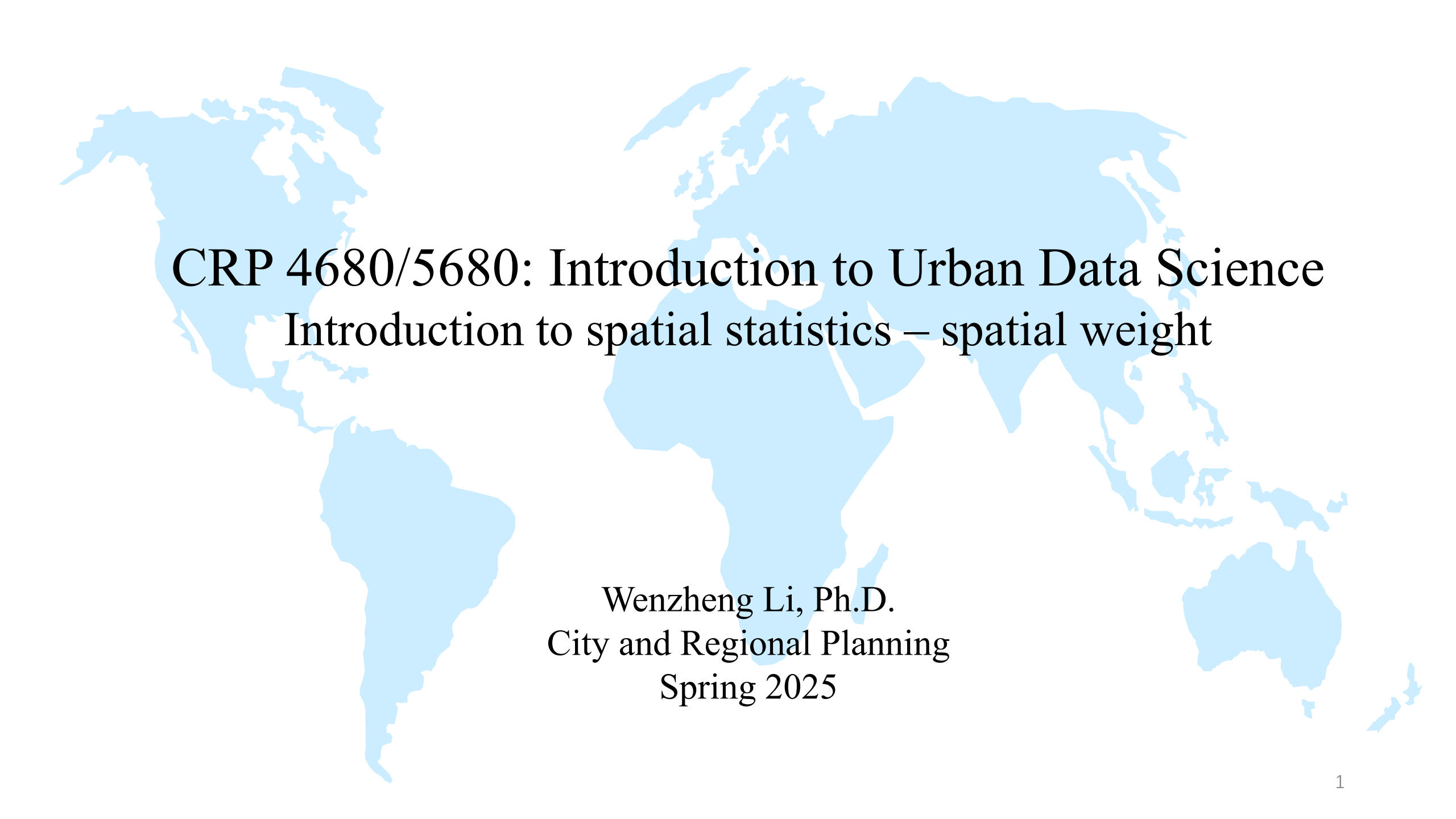




CRP 4680/5680: Introduction to Urban Data Science

Introduction to spatial statistics – spatial weight

Wenzheng Li, Ph.D.
City and Regional Planning
Spring 2025

Final Project Proposal (5% of the total grade)

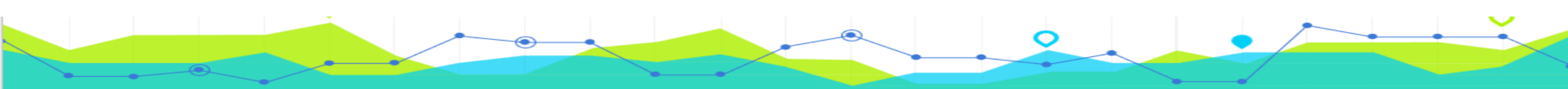
Final project proposal due on **March 28 at 11:59pm**. The final projects can either be individual submissions or submissions of groups of two.

Final project proposal (1- 2 pages)

- A well-defined research question and research background
- For CRP5680, you need to cite five relevant references in your proposal. The references must be from peer-reviewed journals or reports from reputable institutions.
- Possible data sources (with website links if possible)
- Having a clear methodology is desirable. If not, try to propose methods related to this course based on your current knowledge. If you are unsure whether tools exist to achieve your objectives, you can clearly describe your envisioned goals in detail - this helps me understand your research and assist you in finding suitable methods.

Expectations:

- The project should be analytical, not just descriptive.
- Use this final project to assist your exit projects and client-based tasks.



Course Plan

Section 0: Python environment setup and jupyter lab (2 classes)

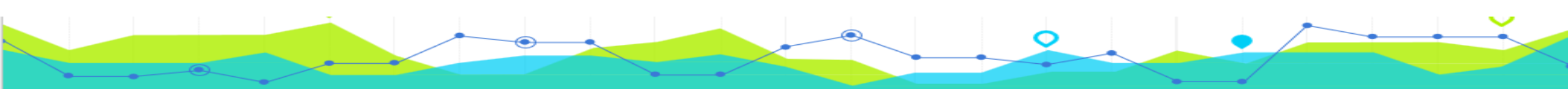
Section 1: Introduction to Python and Data Techniques (11 classes)

- Python basics
- Data management
- Data visualization: aspatial data, spatial visualization, interactive mapping
- Geospatial operations
- Data collection: Map APIs, Census APIs

Section 2: Exploratory Spatial Data Analysis (ESDA) and Spatial Econometrics

- Spatial weights and spatial dependence (2-3 classes)
- Regression and spatial regressions (3-4 classes)

Section 3: Machine Learning (6 classes)



How Do We Determine a Fair House Price?

Scenario:

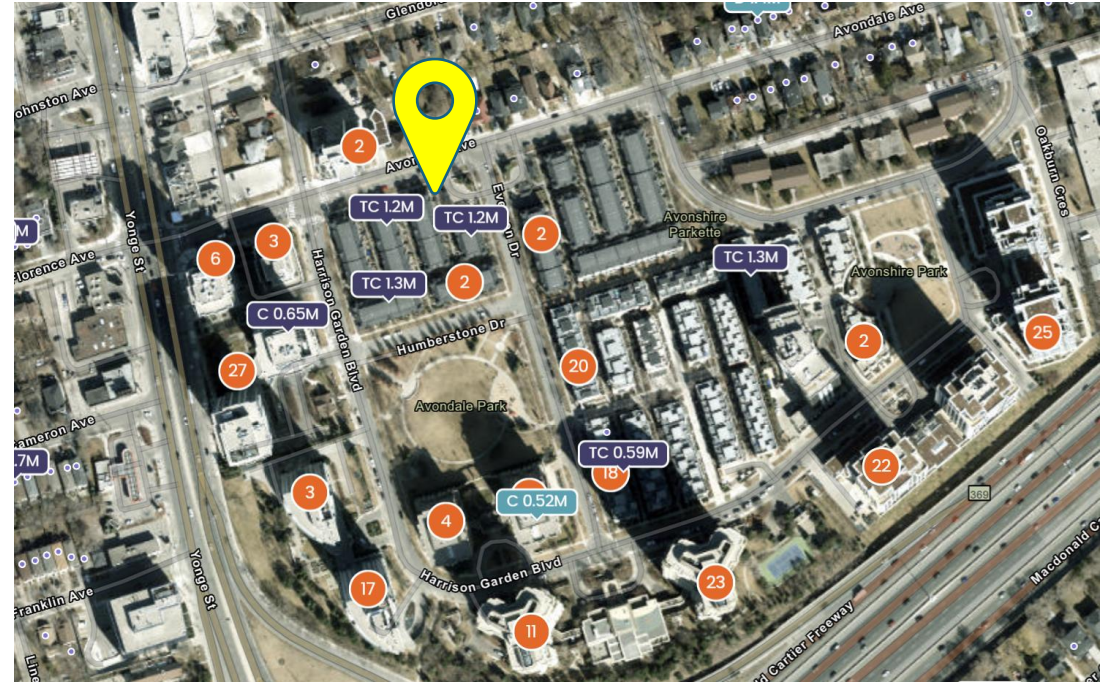
You are looking at a house and considering buying in a nice neighborhood.

Question: What factors can help you determine if the price is reasonable?

- ***Prices of nearby houses***

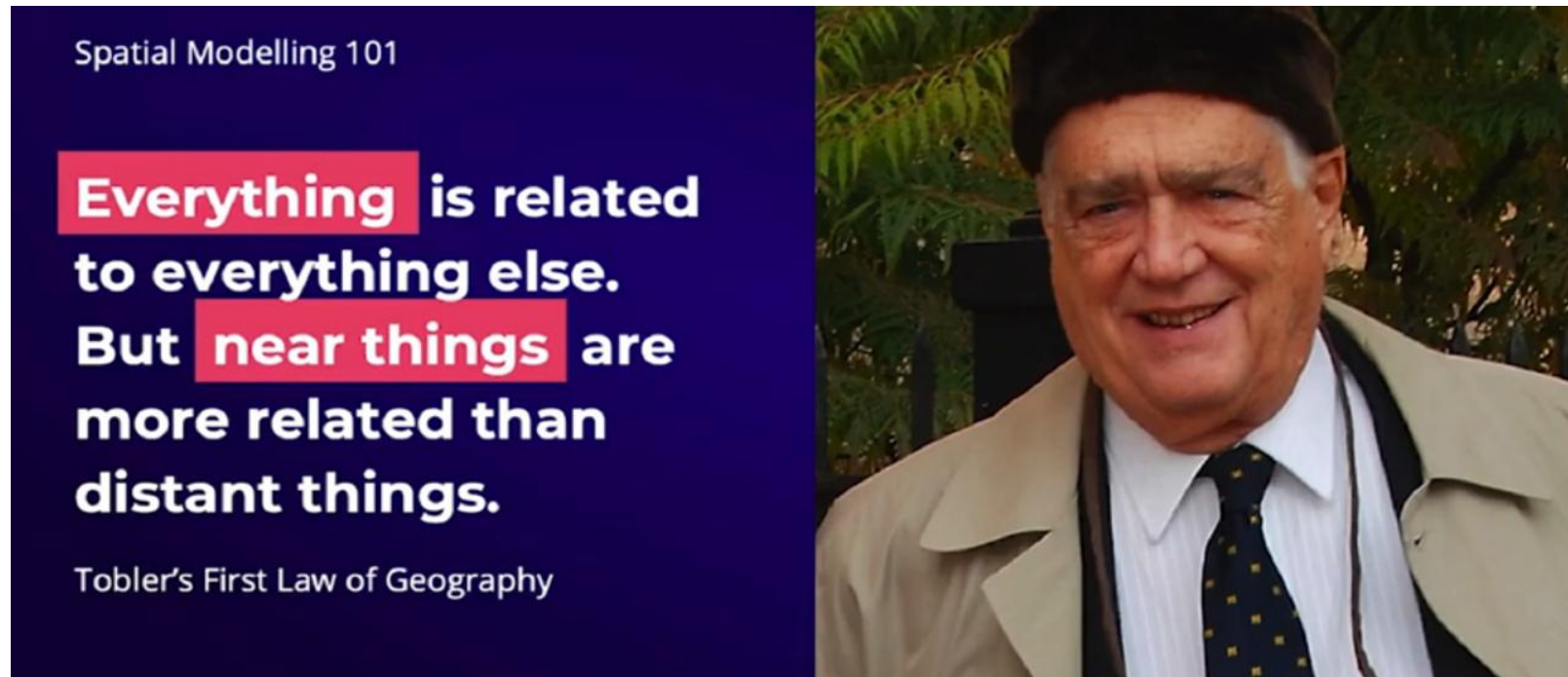
- Historical transaction prices
- House size, layout, and renovations
- Neighborhood amenities, schools, transportation

We reference nearby house prices because properties that are closer together are more similar and have more comparable prices..



Spatial dependence/Spatial autocorrelation

- Geographic data tend to be spatially dependent.
- ***Spatial dependence*** is "the propensity for nearby locations to influence each other and to possess similar attributes" (Goodchild, 1992, p.33).
- **Spatial autocorrelation** is the **statistical measure** of spatial dependence

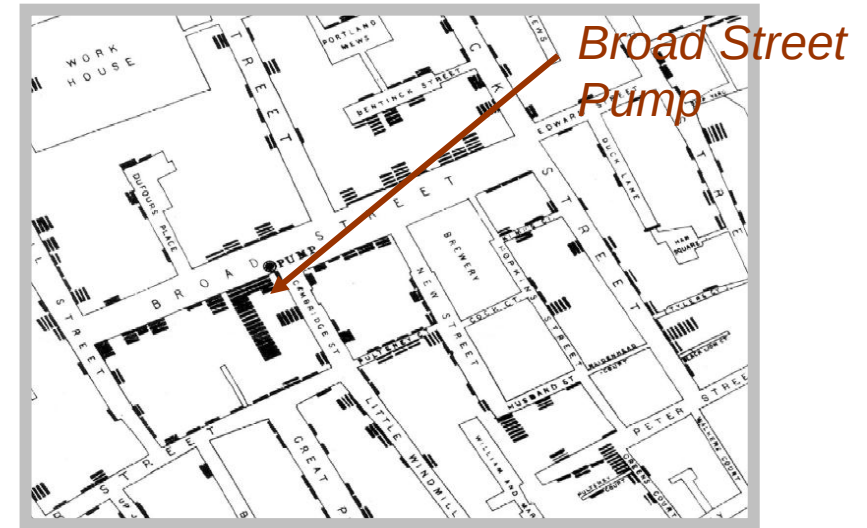
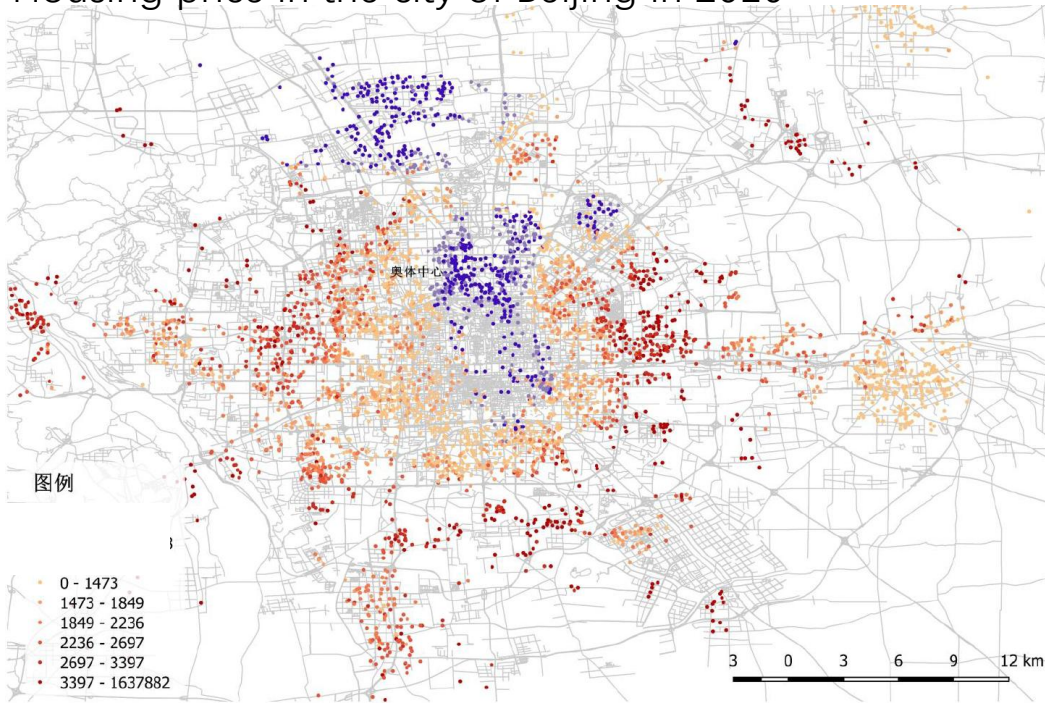


Examples – spatial dependence

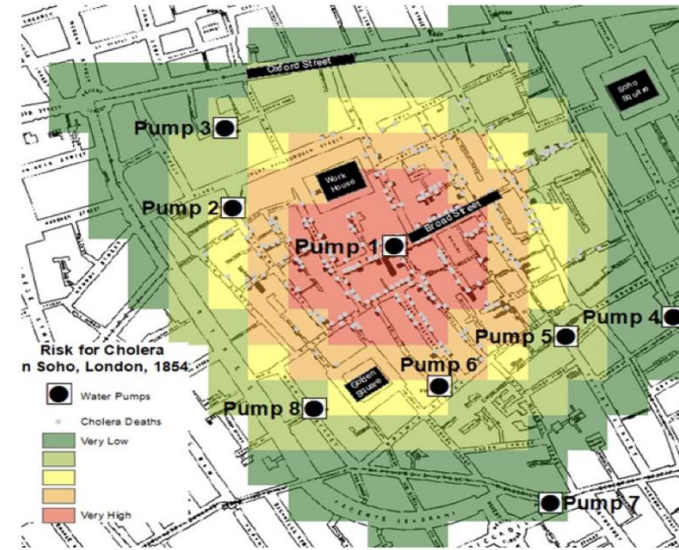
Common in many real-world phenomena, such as:

- House prices in a neighborhood
- Air pollution levels across a city
- Disease spread in epidemiology

Housing price in the city of Beijing in 2016



water pump-wells (●) deaths from cholera (|||||)



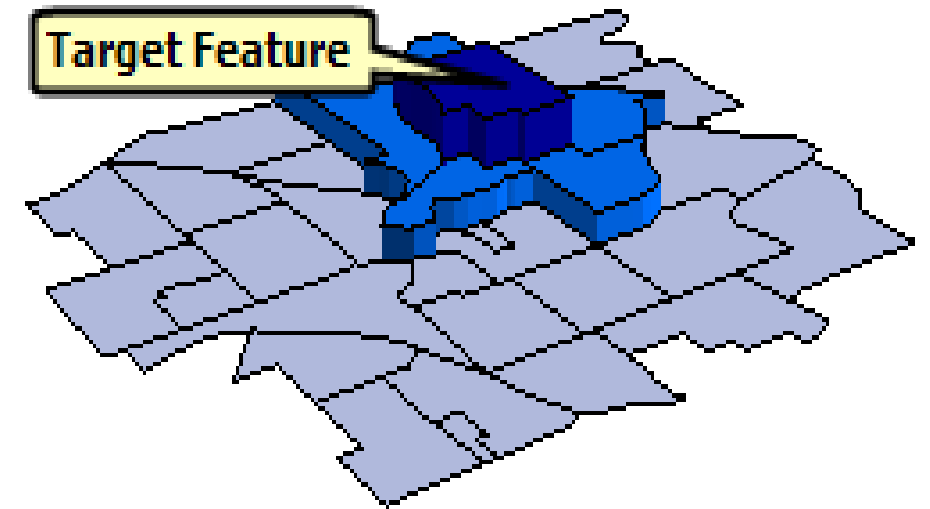
John Snow's map revealed how cholera cases clustered around contaminated water pumps - disease outbreaks were spatially dependent.

Spatial weights?

If nearby properties tend to have similar values, how can we systematically measure this spatial relationship? – we use ***spatial weights***

Spatial weights are representation of the spatial relationship (topology) of your data, guiding each target point to find its neighbors.

- ***Who are my neighbors?***
- ***How important is each neighbor?*** how much weight do we give to our neighbors? This is useful when calculating Spatial Autocorrelation when understanding the similarity between me and my neighbors – we will cover this later



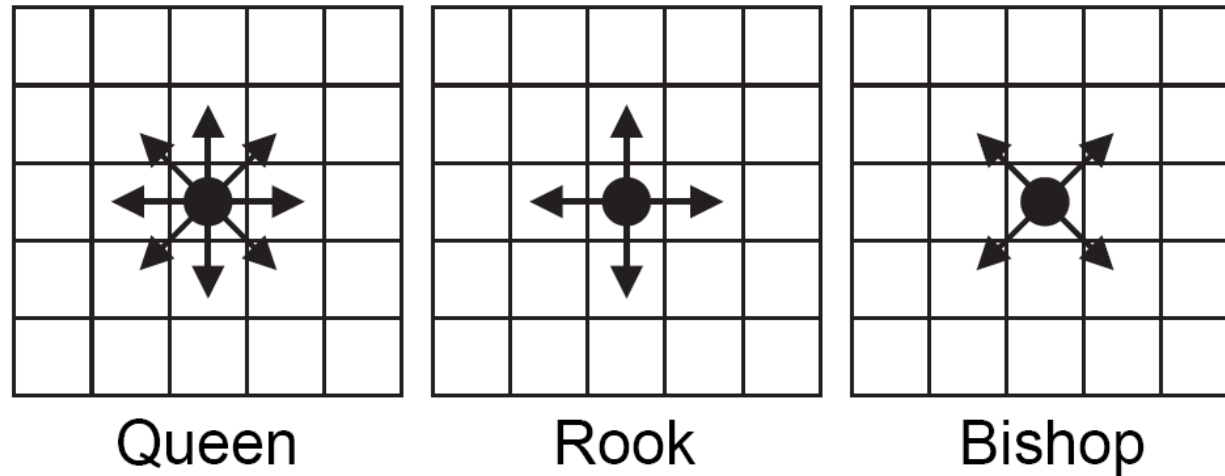
Types of Spatial Weights

- Fundamentally, there are two strategies for creating weights to quantify the relationships among data features:
 - **Contiguity/adjacency weights** – whether a feature is sharing a border with neighbor
 - **Distance weights** – setting a threshold (bandwidth) based on distance or number of neighbors



Spatial weights

Binary: polygon contiguity

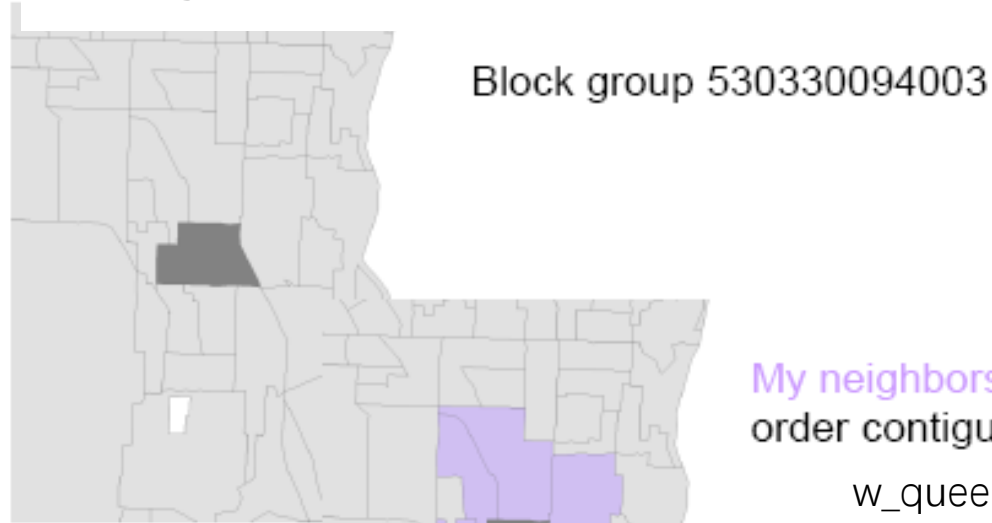


- Binary: If zone j is adjacent to zone i , the interaction receives a weight of 1, otherwise it receives a weight of 0 and is essentially excluded.
- effective when polygons are similar in size and distribution and when spatial relationships are a function of polygon proximity (the idea that if two polygons share a boundary, spatial interaction between them increases)
- you can define a range of contiguity matrices: 1st nearest, 2nd nearest, 3rd nearest, etc.



Spatial weights: Contiguity

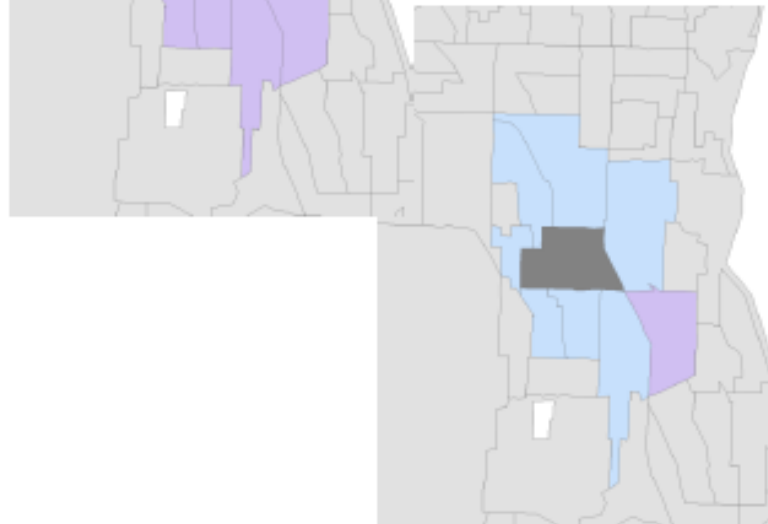
Defining a neighborhood



My neighbors (Queen's 1st
order contiguity

```
w_queen = weights.contiguity.Queen.from_dataframe(df)
```

In ArcGIS: "CONTIGUITY_EDGES_CORNERS"



My neighbors (Rook's 1st
order contiguity

```
w_rook = weights.contiguity.Rook.from_dataframe(df)
```

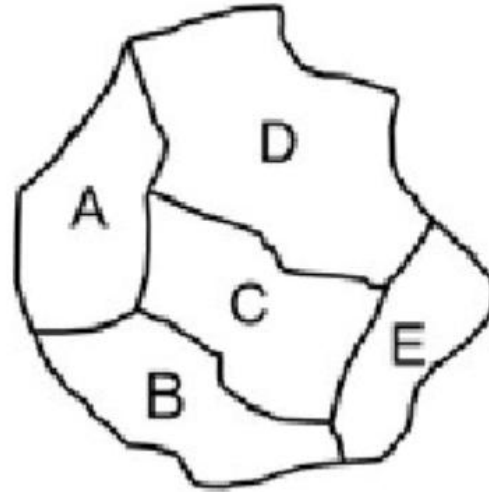
In ArcGIS: "CONTIGUITY_EDGES_ONLY"

Spatial weight matrix – Rook Weight Example

We use a **spatial weight matrix** (W) to store the spatial relationships between multiple spatial entities in a dataset. Mathematically, it is an $n \times n$ matrix, where n is the number of observations

- The diagonal elements w_{ii} are often set to zero (a location does not influence itself).
- Each element w_{ij} represents the spatial relationship between location i and location j .

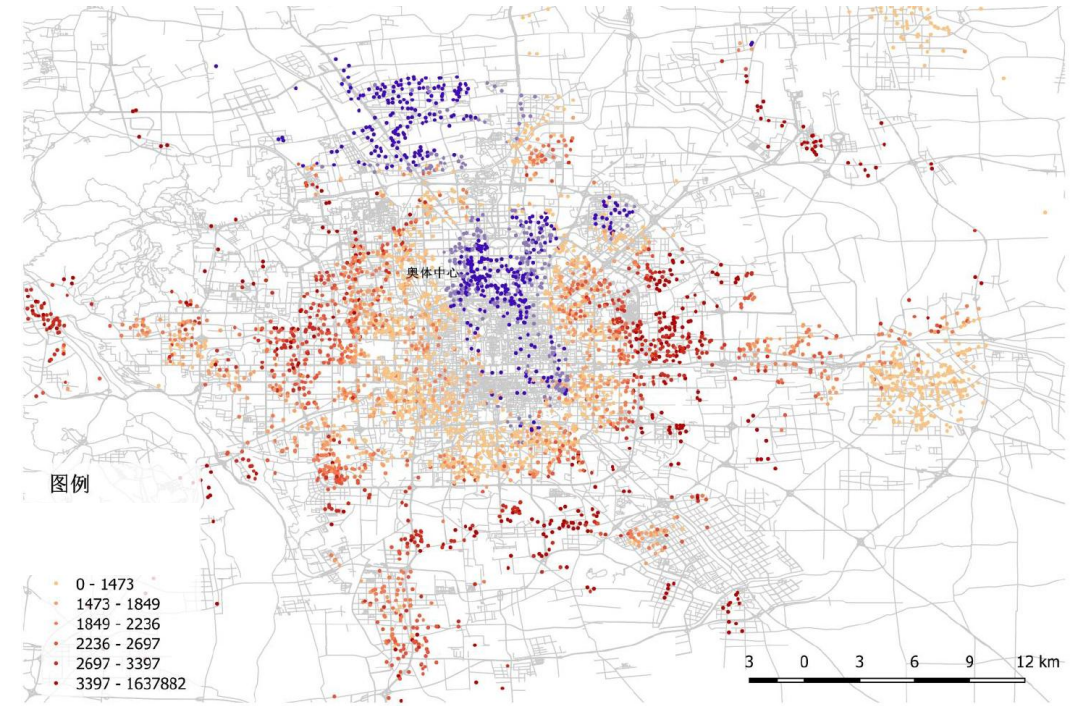
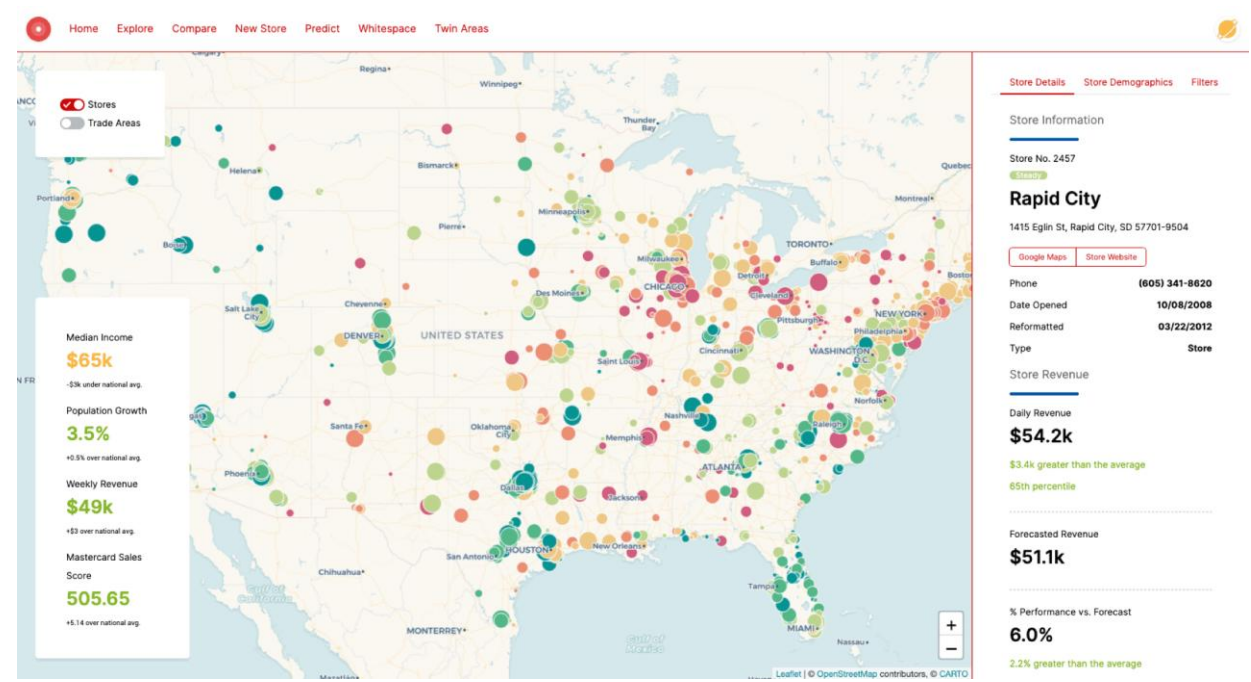
Sample Region and Units



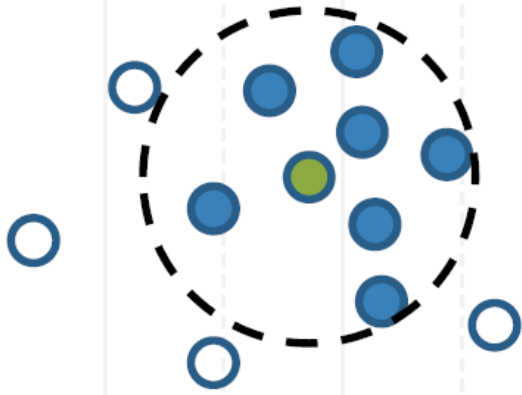
Simple Contiguity (rook) Matrix

	A	B	C	D	E
A	0	1	1	1	0
B	1	0	1	0	1
C	1	1	0	1	1
D	1	0	1	0	1
E	0	1	1	1	0

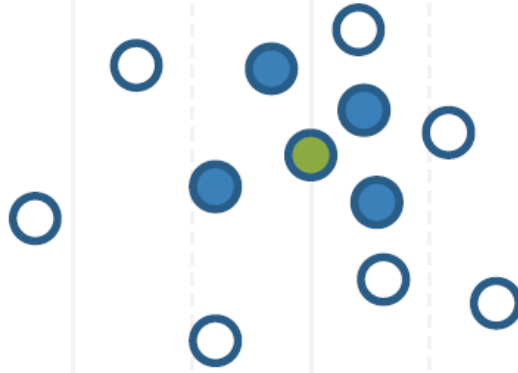
Point data do not have contiguity neighbors.



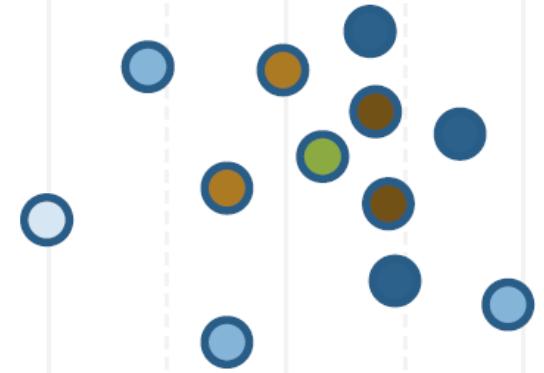
- Point: Distance Based Weights



Distance Threshold



K-Nearest Neighbor



Inverse Distance

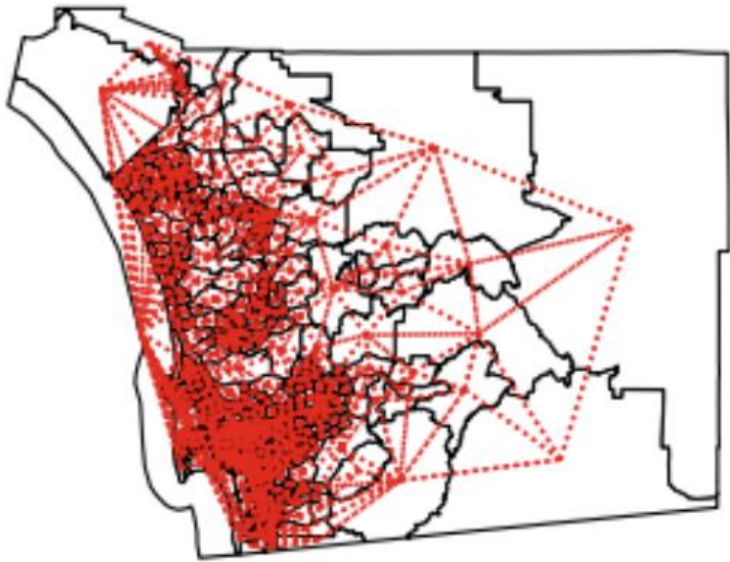


Applying Distance-Based Methods to Polygons

Polygons represent areas rather than discrete points

Convert Polygons to Points (Centroids)

- The centroid (geometric center) of each polygon is used as a representative point.
- Once converted to points, distance-based methods (fixed distance bands, KNN, inverse distance weighting) can be applied.

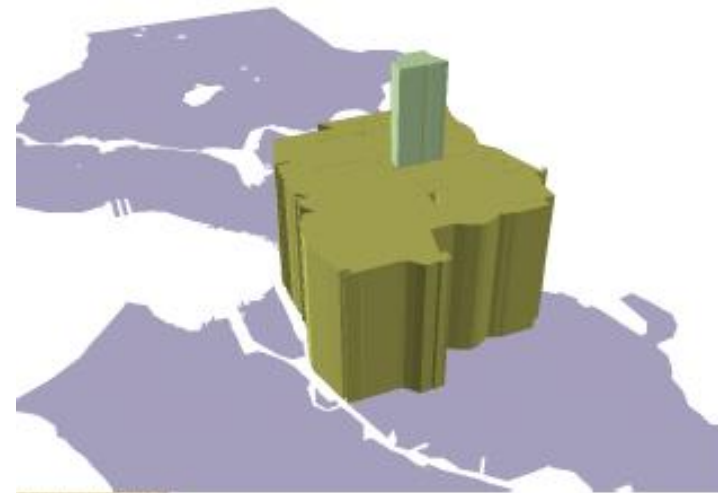
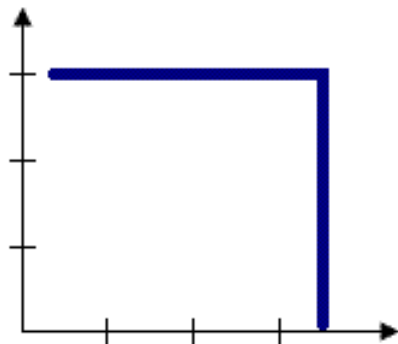


Spatial weights matrix

Binary: fixed distance band

- neighboring features are defined within a certain threshold distance and are **weighted equally**.
- Features outside the specified distance don't influence calculations (their weight is zero).
- If you are studying commuting patterns and know that the average journey to work is 15 miles—you may want to use a 15-mile fixed distance for your analysis.

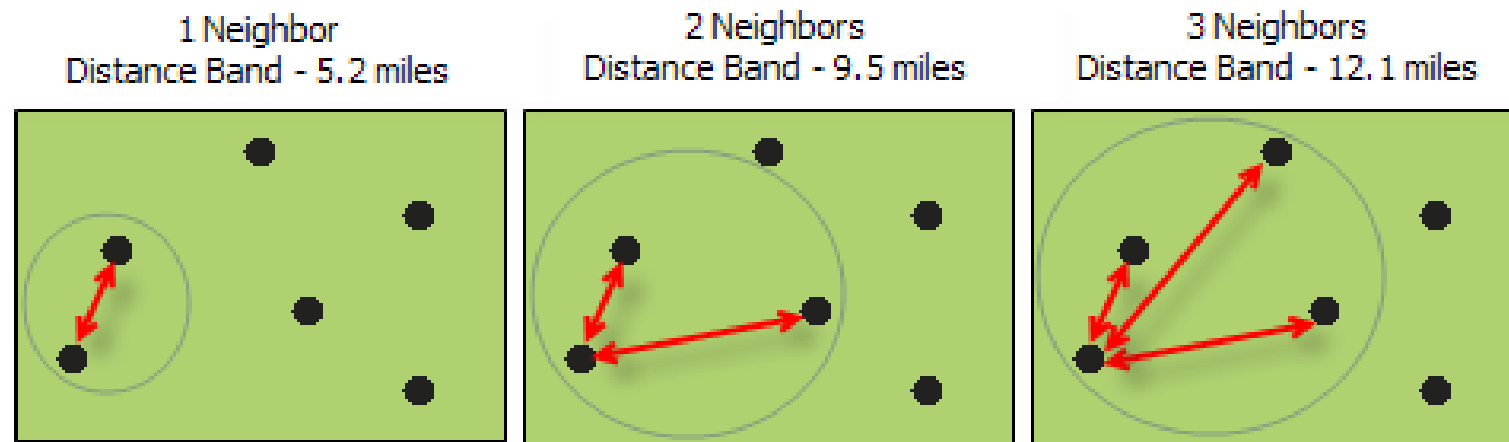
Distance band (sphere of influence)



Spatial weights matrix

Binary: k-nearest neighbors

- Weights based on each feature is assessed within the spatial context of a fixed number of its closest neighbors.
 - In locations where feature density is high, the spatial context of the analysis will be smaller.
 - This method ensures there will be some neighbors for every target feature



Kernel Weights - Beyond Binary Weights

Limitation of Binary Weights:

"Binary weights oversimplify spatial relationships by treating all neighbors equally."

According to Tobler's First Law, nearby locations are more related, but binary weights fail to reflect this gradual relationship.

Kernel weights provide a more realistic spatial representation by assigning **gradual** influence based on distance, allowing smoother spatial analysis."

Tobler's First Law of Geography:
"everything is related to everything else,
but near things are more related than
distant things"

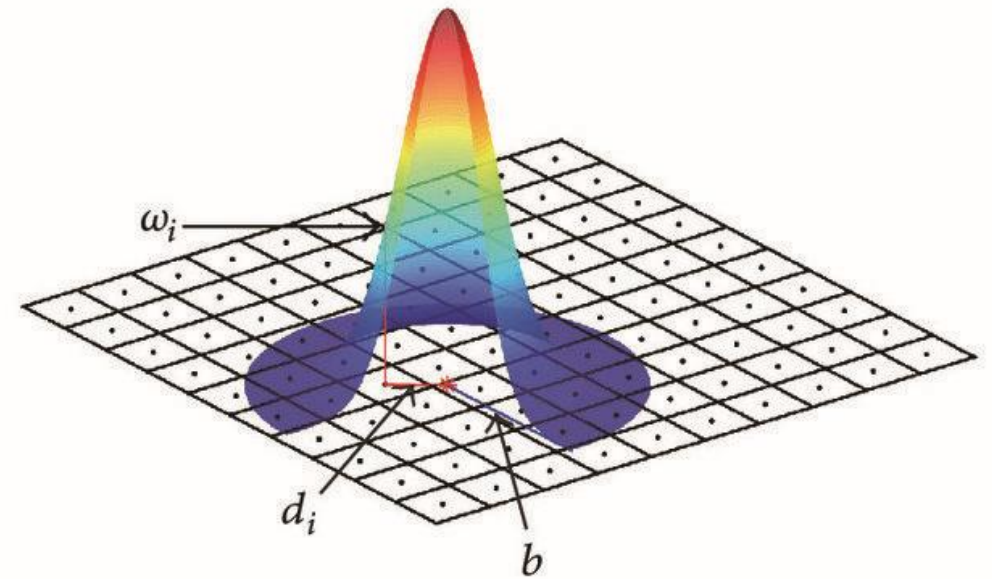
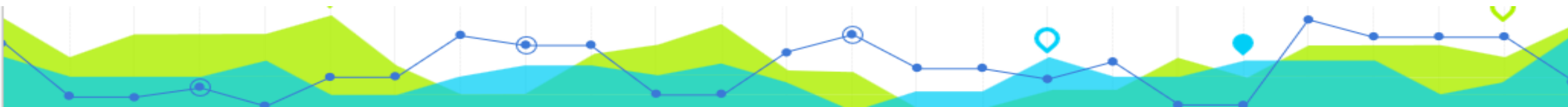


Figure Assumes Triangular Kernel

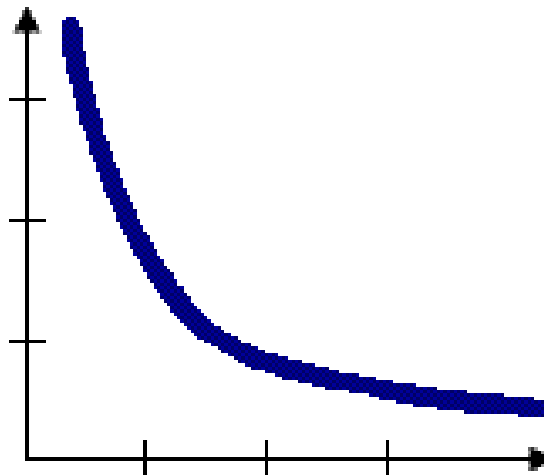
Other kernel weights



Spatial weights matrix

Kernel weights: Inverse distance weighting

- All features impact/influence all other features, but the farther away something is, the smaller the impact it has
- Appropriate with continuous data or to model processes where the closer two features are in space, the more likely they are to interact/influence each other.
- You can specify a threshold distance: Features outside the specified cutoff are ignored; You can force all features to be a neighbor of all other features by setting Threshold Distance to zero (large number of computations!).
- Suitable cases:
 - Air Pollution
 - Temperature & Rainfall
 - Real Estate & Housing
 - Disease Spread
- assumes that the underlying data varies gradually and smoothly across space.

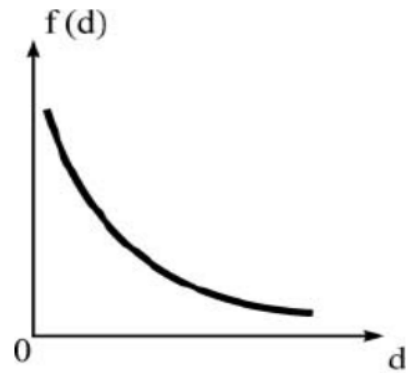


Neighbors are weighted based on the inverse of the log of their distance from my block group's centroid



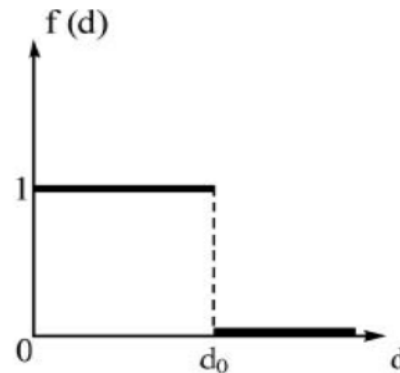
Different spatial weighting functions that define how weights, $f(d)$, decrease with increasing distance (d)

Continuous Distance Decay



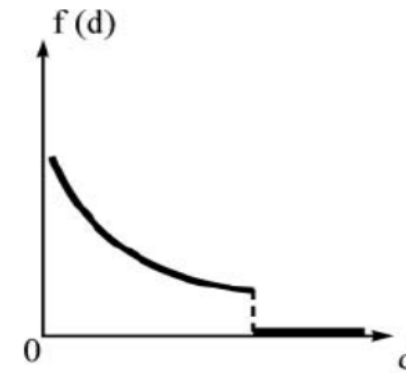
1a

Binary and Stepwise Distance Decay

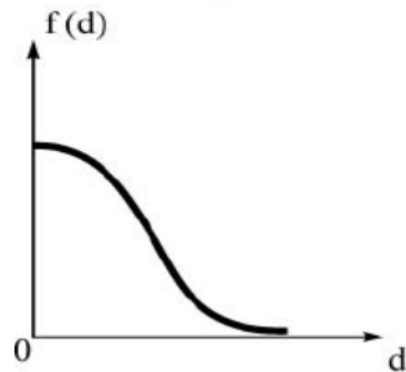


2a

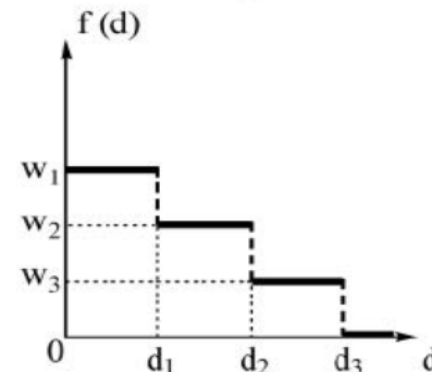
Hybrid Distance Decay



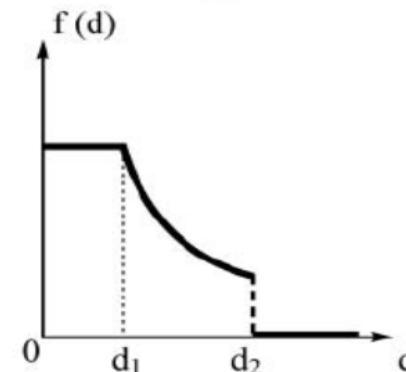
3a



1b



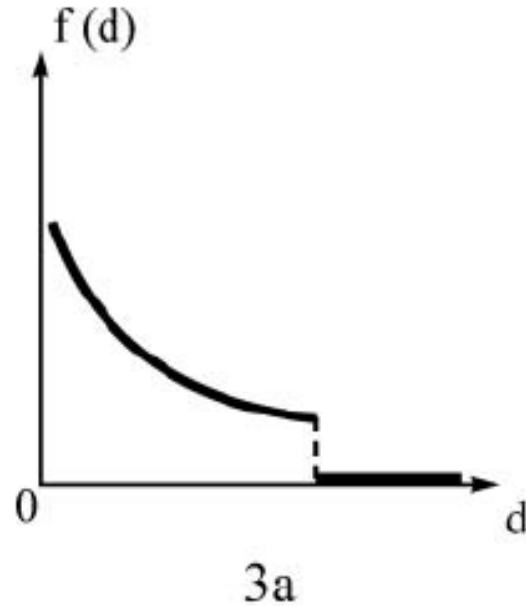
2h



3h

In this week's lab, we will cover the following spatial weights strategies:

(1) Distance Based: *Kernel + fixed distance bandwidth*



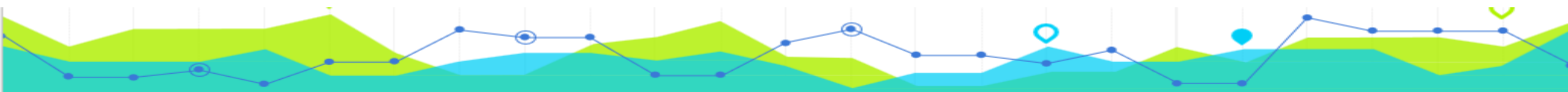
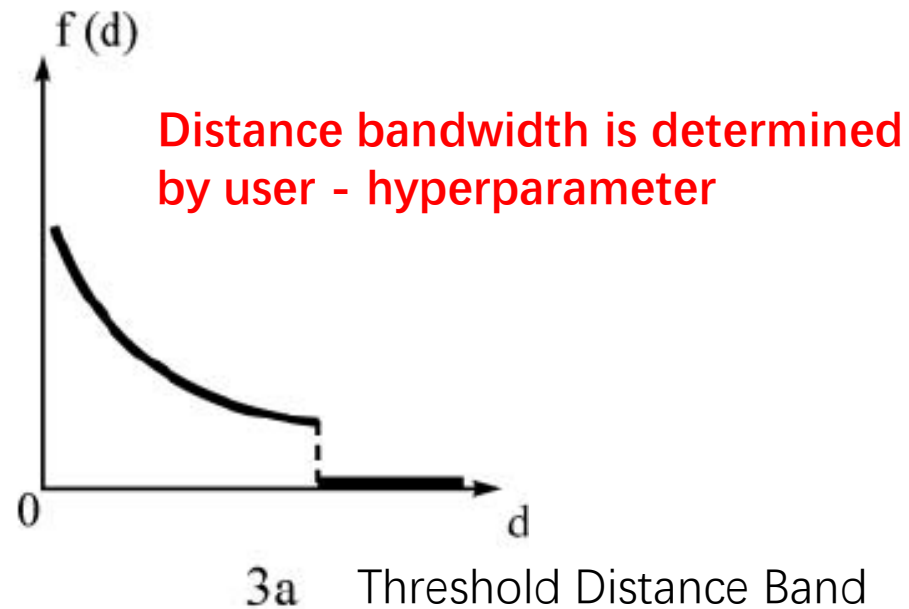
Fixed distance bandwidth is determined by python based on maximum knn=2



(2) Distance Based: *Adaptive Bandwidths*

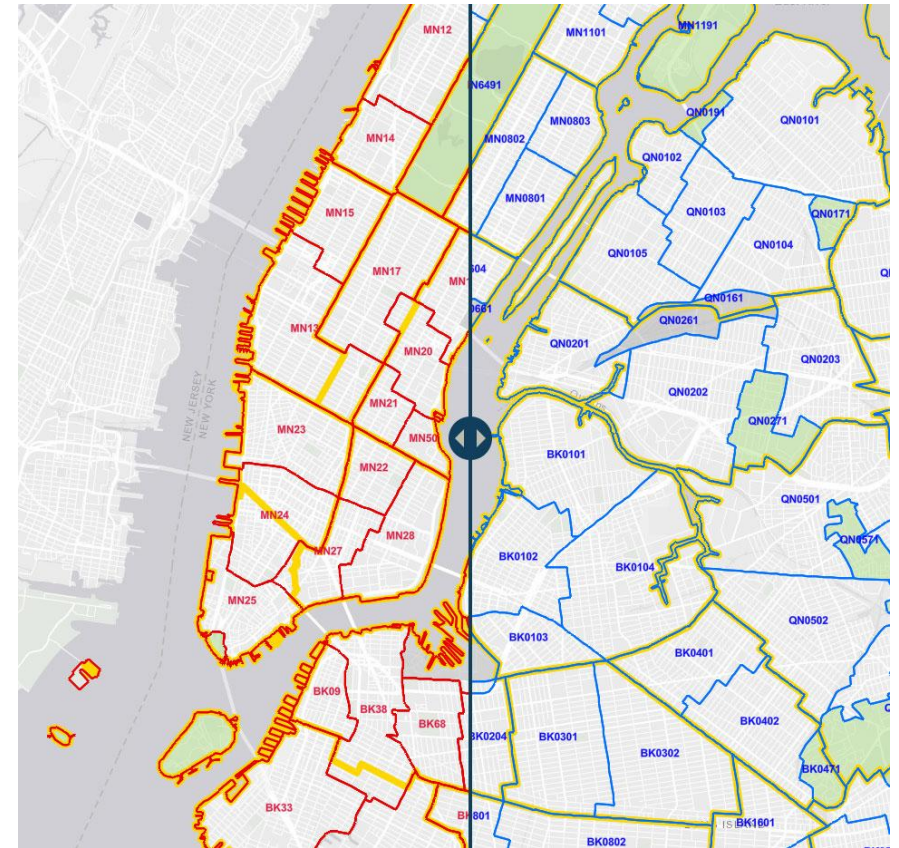
- Some regions have differing densities, so the number of neighbors may differ significantly
- The bandwidth distance for each observation varies dynamically based on the distance required to include exactly k nearest neighbors.

(3) Distance Based: *Fixed distance band + kernel (IDW)*



(4) Binary Distance Based - Block Weights

- Similar to clustering standard errors – observations that are within the same larger unit (i.e. administrative boundaries – census tracts in zipcodes, counties in a state, nations in a continent) are considered neighbors



(source: NYC planning)

Summary - Understanding Spatial Weights

- Step 1: Defining Neighbor Selection Methods - determine how neighbors are selected: e.g., Fixed Distance Band, K-Nearest Neighbors (KNN), Contiguity-Based (Adjacency).
- Step 2: Assigning Weights to Neighbors - Once neighbors are determined, we assign weights based on their relationship. E.g., Binary Weights (0,1), Distance decay kernel functions



What types of spatial weights to choose?

- you should select a conceptualization that best reflects how features actually interact with each other (giving thought, of course, to what it is you are trying to measure). Theory should inform this decision!
- If you are measuring clustering of a particular species of seed-propagating tree in a forest, for example, some form of inverse distance is probably most appropriate. If you are measuring social processes at the neighborhood level, polygon contiguity is a good choice



Distance Methods

Specifies how distances are calculated from each feature to neighboring features.

- EUCLIDEAN DISTANCE: The straight-line distance between two points (as the crow flies)
- MANHATTAN DISTANCE: The distance between two points measured along axes at right angles (city block); calculated by summing the (absolute) difference between the x- and y-coordinates

